

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

I P. Sathank C 192425944 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sathank C

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1	1	1	1
Accuracy	10%				
Clarity	5%				

Total Marks 28 /40

Signature of the Faculty [Signature]

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and

the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

①

File size = 150 MB

Band width = 50 mbps

Segment size = 1500 bytes, Datalink overhead = 40

Step 1:- Number of segments and frames

Total file size = 150 MB = 150,000,000 bytes

Number of segments = $\frac{\text{Total bytes}}{\text{segment size}}$
 $= \frac{150,000,000}{1500} = 100,000 \text{ segments}$

Number of frames = 100,000 frames

Step 2:- Datalink layer overhead.

Overhead per frame = 40 bytes

Total overhead = $100,000 \times 40 = 4,000,000$

Total data transmitted = $150,000,000 + 4,000,000$
 $= 154,000,000 \text{ bytes}$

In bits = $154,000,000 \times 8 = 1,232,000,000 \text{ bits}$

Transmission time = $\frac{\text{Total bits}}{\text{Band width}}$

$= \frac{1,232,000,000}{50,000,000}$

$= 24.64 \text{ seconds}$

Datalink layer frames each segment with header / trailer that carry addressing and a CRC / check sum for error detection. on a faulty link it uses ARQ so each hop delivers frames free of transmission errors.

② given,
sliding window size = 6 frame, frame size = 1024 bytes
Total frame sent = 48

Number of transmission windows

$$\text{number of windows} = \text{Total frame} \div \text{window size} \\ = 48 \div 6 = 8$$

Retransmission

one lost frame per window $\times 8$ windows = 8 retransmission

- Flow control let the sender transmit multiple frame before waiting for an acknowledgement, instead of stopping after every single frame.
- This keeps the transmission link busier, improves throughput and still protects a slower receiver from being overwhelmed since the window size is bounded by the receiver buffer capacity.

Number of windows = 8

Total retransmission = 8 frames

③ IP packet size = 12,000 bytes

MTU = 1500 bytes, IP header = 20 bytes.

Usable data per fragment

Data payload per fragment = MTU - header = 1500 - 20 = 1480

Number of fragments

Number of fragments = $\text{ceil}(12,000 \div 1480) = \text{ceil}(8.11) = 9$ frags

8 fragments carry 1480 bytes each = 11,840 bytes, the 9th fragment carries the remaining 160 bytes

Total overhead = $9 \times 20 = 180$ bytes.

→ The network layer (IP) fragments packets larger than a link's MTU using the Identification field, the Fragment Offset field and the Max-Fragment Flag.

→ The destination host uses these fields to reassemble all fragments into the original packet before passing it up to the Transport layer.

Number of fragments = 9

Total header overhead = 180 bytes.

④

Video = 600 MB

Session layer = 60 MB

Failure occurs after 390 MB

Check points created before failure

Check points occur at 60, 120, 180, 240, 300, 360 MB

Number of check points before failure = Floor $(\frac{390}{60})$
= 6 Check points

Data to be retransmitted after recovery

Last check point reached = 360 MB

Data lost / retransmitted = $390 - 360 = 30$ MB

→ The session layer inserts synchronization check points to mark safe recovery points during a long transfer.

→ If a failure occurs, the session can resume from the last check point. Instead of resending the entire 600 MB transfer, saving significant time and bandwidth - here only 30 MB needs to be resent instead of 390 MB.

Check points before failure = 6

Data to be retransmitted = 30 MB